

Vitaly Kurnosenko

DataOps / Big Data Engineer | Kubernetes & Spark | Data Lake | CI/CD | On-prem AI

Place of Residence: Asia (open to relocation to Limassol, Cyprus) | **Phone / Skype:** +86 15601694273 | **Email:** rikkimortycrypt@gmail.com | **WeChat:** porohnya | **Telegram:** @vitaly_kur | **Languages:** English - B2 / Upper-Intermediate, fluent spoken and written; Chinese - HSK 4; Russian - native

Professional Summary

DataOps / Big Data engineer with hands-on experience operating Kubernetes clusters (GKE and **K3S/Rancher** on bare-metal EC2), running **Spark on Kubernetes**, building Data Lakes on object storage (S3, GCS, Yandex Object Storage), and deploying **on-premise AI infrastructure**. Cover the full lifecycle - capacity planning, networking, security, CI/CD, monitoring, and disaster recovery.

5+ years on production data platforms in international fintech / e-commerce products (Yofi, USA - anti-fraud for Shopify / Lululemon; Sinoptics AI - on-prem AI for invoice / document processing). Strong in Python and SQL, comfortable maintaining Java/Scala Spark jobs and reading/patching Go services. Linux is my primary production environment.

Core strengths: physical implementation of Big Data environments, Spark/Hadoop performance tuning, security-by-design (Workload Identity, Secrets Manager, RBAC), incident management, and SLO-grade operational discipline.

Core Skills (mapped to the role)

Area	Tools and experience
Containers / Orchestration	Docker, containerd, Kubernetes, GKE, K3S (Rancher) on bare-metal AWS EC2, Helm, KubernetesPodOperator, Workload Identity, namespaces, RBAC, image pull secrets
Big Data & Spark	PySpark on Dataproc Serverless / Spark on K8s , custom Debian-based Spark Docker images, Hudi, Delta Lake, BigLake, Hadoop FS, partitioning, performance tuning
Data Lake / Object Storage	S3 (AWS, incl. China region), GCS, Yandex Object Storage, BigLake, lakehouse patterns (Hudi/Delta), Trino-class engines (BigQuery, Athena)
Languages	Python (PySpark, FastAPI, Airflow, boto3, k8s client), SQL (PostgreSQL, BigQuery, Spanner, MSSQL, Trino-style), JavaScript/TypeScript; read and patch Java/Scala Spark jobs and Go services
Linux & networking	Ubuntu/Debian, systemd, bash, nginx, certbot/Let's Encrypt, UFW, IPtables, DNS, TLS, jemalloc tuning
CI/CD & Automation	GitHub Actions , AWS CodeCommit + CodeBuild , Google Cloud Build, Pulumi (IaC) , bash deployment scripts in the spirit of Ansible/Chef for PostgreSQL and stack installers, Docker BuildKit
Data orchestration	Airflow (KubernetesPodOperator pattern), Airbyte, DBT, Hatchet, n8n

Security & DR	AWS Secrets Manager, GCP Secret Manager, IAM / Workload Identity, RBAC, encryption at-rest / in-transit, automated PostgreSQL backups with retention, Slack alerting on DAG failures
Monitoring & Alerting	Cloud Logging / Monitoring, Slack webhooks, structured logs, runbook-driven incident response
AI infrastructure	On-prem Dify (LLM platform) on Ubuntu VM with Docker Compose, Hatchet workflow engine on K8s, OCR / LLM pipelines, retrieval / agent context

Professional Experience

DataOps / AI Platform Engineer - Sinoptics AI

March 2025 - October 2025 (remote)

- Deployed and operate an **on-premise AI stack** for document (invoice) processing: Dify v1.10 on Ubuntu VM in Yandex Cloud + PostgreSQL, isolated Docker Compose stack, nginx as edge with automatic Let's Encrypt issuance and rotation via certbot.
- Migrated a FastAPI + Celery / Redis microservice onto a **K3S cluster (Rancher)** on bare-metal AWS EC2 (China region, cn-northwest-1); built a hybrid CI/CD: GitHub Actions → AWS CodeCommit → CodeBuild → ECR → `kubectl rollout`, with `kubeconfig` stored in AWS Secrets Manager.
- Stood up a **Hatchet workflow engine on K8s** (namespace, ingress, external secrets) to orchestrate long-running OCR / LLM tasks; configured pod monitoring and alerting.
- Implemented automated PostgreSQL deployment and backup on Ubuntu (bash scripts in the style of an Ansible role: package install, `pg_hba.conf` / `postgresql.conf` tuning, UFW, 7-day retention backups, cron).
- Designed a layered document-processing architecture (n8n → specialized AI agents → aggregator → report) with a security model (RBAC, encryption at-rest/in-transit, audit logs, GDPR / Chinese data law compliance).

Data Engineer - Yofi Inc. (USA)

February 2022 - October 2025

Yofi is an anti-fraud / customer-clustering platform for Shopify merchants (Lululemon and others), where I owned the DataOps part of the Big Data stack.

- Operated **GKE clusters** (yofi-hub) across dev/prod GCP projects: configured Workload Identity, image pull secrets for Google Artifact Registry, namespace separation (hub-dev / hub-prod), roles/rolebindings via `kubectl apply`, context switching, and `kubeconfig` management.
- Ran **Spark on Kubernetes** via the `KubernetesPodOperator` pattern in Airflow + Dataproc Serverless: built custom **Debian 11/12 Docker images** for Spark with Hudi, Delta Lake, MongoDB connector, BigQuery jar, jemalloc memory tuning, conda/mamba and a scientific Python stack.
- Built a Data Lake layer on **GCS + BigLake + Hudi/Delta** (`raffles_to_hudi`, `mongodb_to_biglake`, `firestore_to_bq`, `spanner_to_bigquery`, analytics / sessions / stats pipelines); tuned partitioning, IO throughput, and shuffle parameters for jobs spanning 1B+ events.
- Maintained **Airflow** (~25 production DAGs): a `create_kpo_task` wrapper around `KubernetesPodOperator`, Slack alerting on failures, env vars, retry policies, `TriggerRule.ALL_DONE` for resilient pipelines; CI for pod images via **Google Cloud Build** to Artifact Registry with `:latest` and `:${SHORT_SHA}` tags.

- Owned **infrastructure-as-code** via **Pulumi** (Python) for the base GCP estate: BigQuery, GCS, Cloud Functions, IAM, dev/prod stacks.
- Integrated the AWS side of the platform: dozens of Lambdas (Node.js / Python) behind API Gateway, SQS/SNS, EventBridge, RDS, Cognito, Neptune, Aurora; CDK batch refresh counters; CloudFormation/CDK for base VPC / Cognito / Secrets / EC2 resources.
- Implemented **disaster recovery & security**: Secrets Manager, encryption, GCS versioning, idempotent DAGs with persisted state in Redis, Slack alerting and runbooks; participated in incident response for ingestion-pipeline failures.
- Provided **DataOps consulting** to product teams: API documentation in GitBook, KPO task templates, training on DBT, Airbyte, and Spark best practices.

Stack: Python, SQL, JavaScript, PySpark, Spark, Hudi, Delta Lake, Airflow, Airbyte, DBT, Kubernetes (GKE), Docker, Pulumi, GCP (BigQuery, GCS, BigLake, Spanner, Pub/Sub, Cloud Build, Artifact Registry, IAM / Workload Identity), AWS (S3, Lambda, API Gateway, SQS/SNS, RDS, Secrets Manager, EC2, ECR, CodeBuild, CodeCommit), MongoDB, PostgreSQL, Neo4j, Redis, Slack alerting.

Head of IT and Finance - Engineering Solutions LLC

March 2013 - December 2017

- Led IT and finance functions; oversaw rollout and operation of ERP, reporting systems, and office IT infrastructure.
- Supported cross-functional decision-making through systems thinking, governance, and process automation.

Head of IT Department - New Engineering Solution

April 2003 - February 2013

- Owned ERP, corporate IT infrastructure, and reporting systems; introduced practices around structured data management and automation.

Relevant Projects (open-source / personal)

- **k3s-fastapi-app** - production template for FastAPI + Celery + Postgres + Redis on K3S / AWS EC2 with hybrid CI/CD (GitHub Actions → CodeCommit → CodeBuild → ECR → K3S).
- **diffy-vm-ubuntu** - one-shot installer for on-prem Dify (Multi-Database Era) on Ubuntu 22.04+ with Postgres / MySQL / OceanBase support, nginx, and certbot.
- **postgresql-yandex-vm** - bash automation (Ansible-role style) for the full PostgreSQL deployment lifecycle on Ubuntu: packages, pg_hba / postgresql.conf, UFW, retention-aware backups, monitoring.
- **yofi-airflow-kubernetes-operator** - KubernetesPodOperator pattern for Spark/Python jobs on GKE with Workload Identity, Cloud Build pipeline, and dev/prod separation.

Education

- **RPANEP** - MBA, Information Technology (*September 2005 - May 2007*)
 - **Southern Federal University** (formerly Rostov State University) - PhD, Mathematical Sciences (*September 1994 - May 2003*)
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Certifications and additional training

- **Certified Accountant Practitioner (CAP)**
 - Advanced training: **Financial accounting and reporting (IFRS), Management accounting**
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Additional Information

- Open to relocation to Limassol, Cyprus; experienced living and working across multiple countries (Russia, China, Thailand) and cross-cultural teams.
- Strong focus on security best practices, disaster recovery, and incident management in business-critical environments (anti-fraud, finance).
- Continuously follow Big Data / AI ecosystem developments and use generative AI tools to accelerate operational tasks and runbook authoring.